

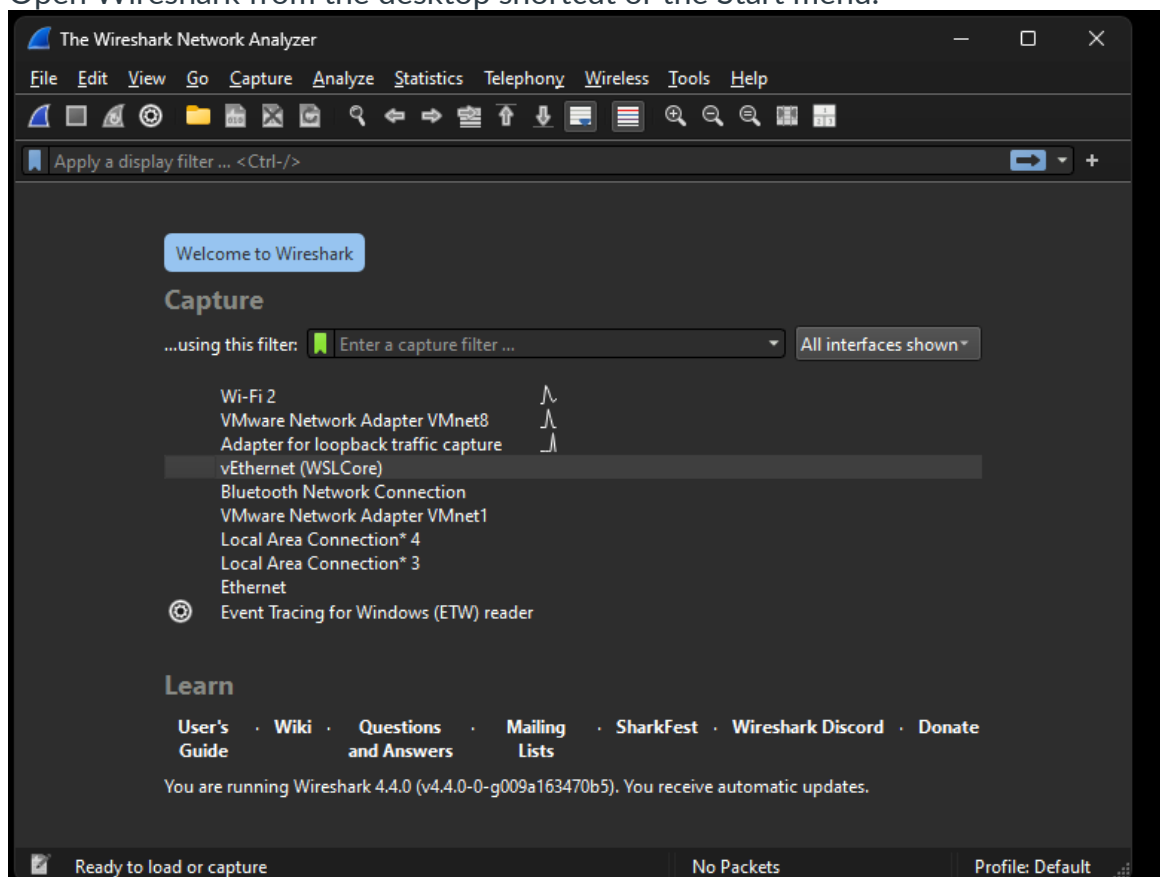
Wireshark Packet Analyzer Setup

Step 1: Download and Install Wireshark:

1. Visit the official Wireshark website (<https://www.wireshark.org/> [Links to an external site.](#)) and download the appropriate version for your operating system.
2. Run the installer and follow the on-screen instructions to complete the installation.

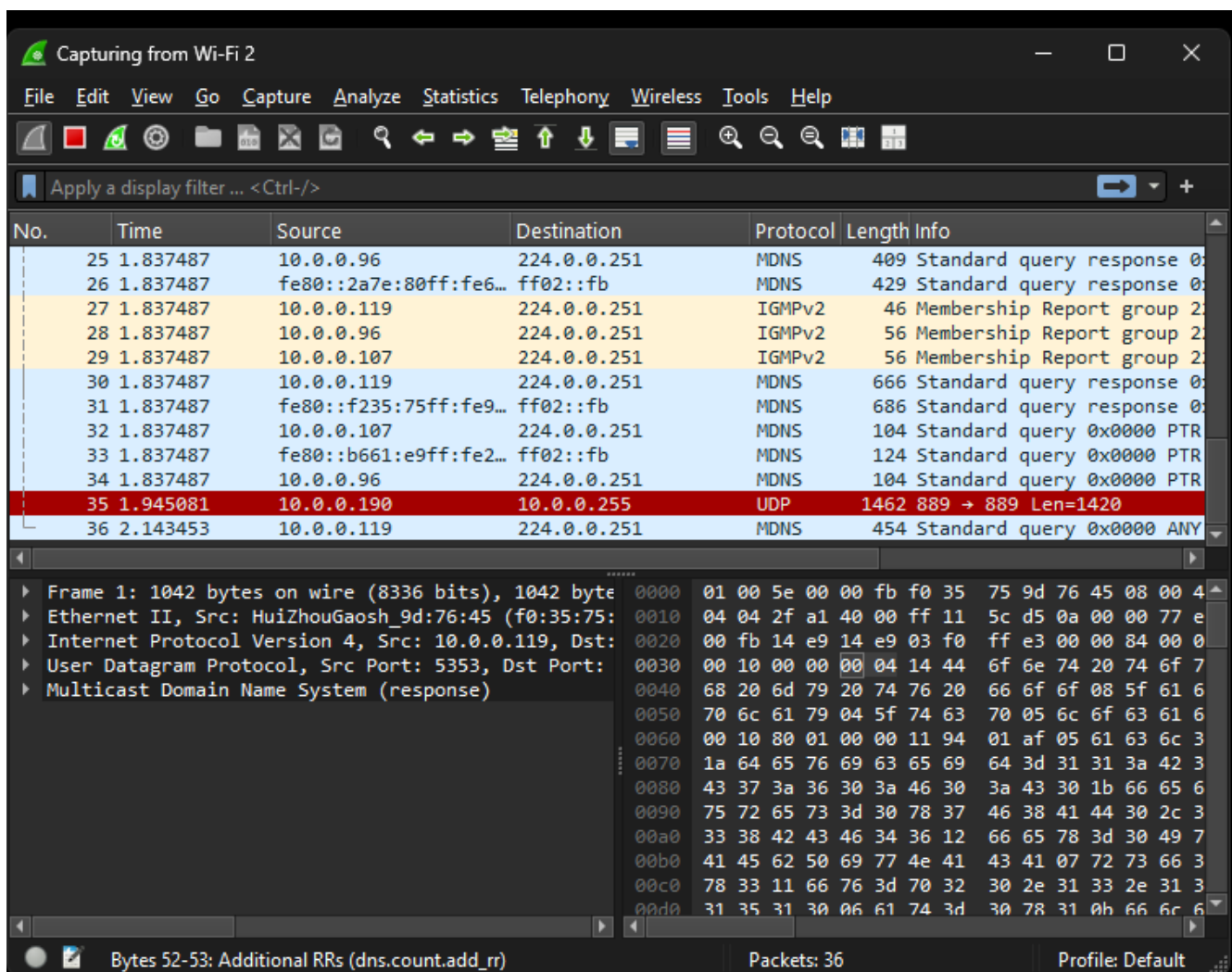
Step 2: Launch Wireshark: (SCREEN SHOT)

1. Open Wireshark from the desktop shortcut or the Start menu.



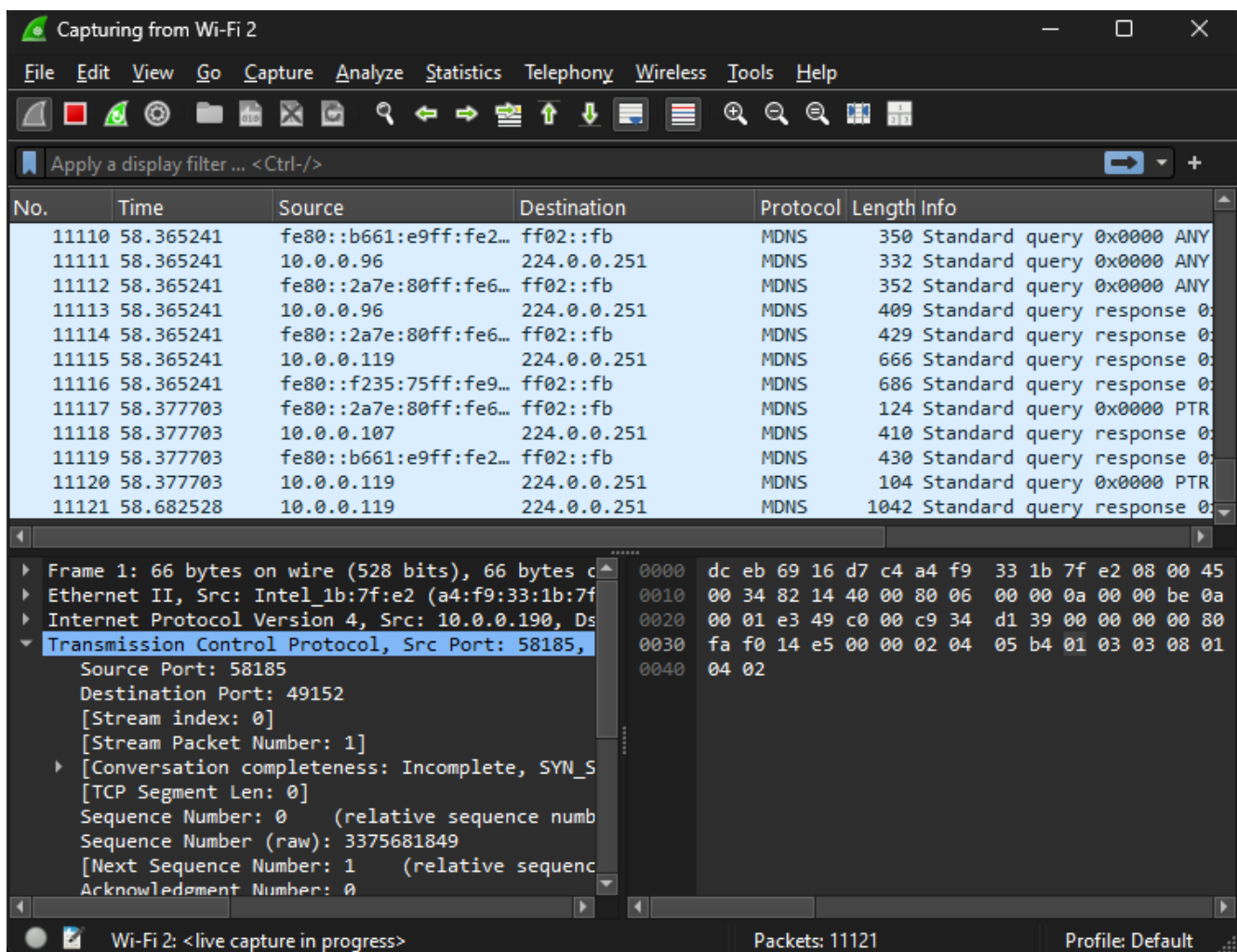
Step 3: Select a Network Interface: (SCREEN SHOT)

- Upon launching, Wireshark will display a list of available network interfaces. Choose the interface that you want to capture traffic from (e.g., Wi-Fi, Ethernet). **Note: this is in a lab environment so the connection will be Ethernet.**
- Click on the desired interface and then click the "Start" button to begin capturing traffic.



Step 4: Capture Packets: (SCREEN SHOT)

- Wireshark will start capturing packets in real-time from the selected interface. The captured packets will be displayed in the main window.
- Observe the different columns that show packet information such as source and destination addresses, protocol, length, etc.
- Let run for about 3 to 4 minutes, while running open a browser and pick three webpages to open. (SCREEN SHOT)
- Then select the red square at the top to stop packet capture.



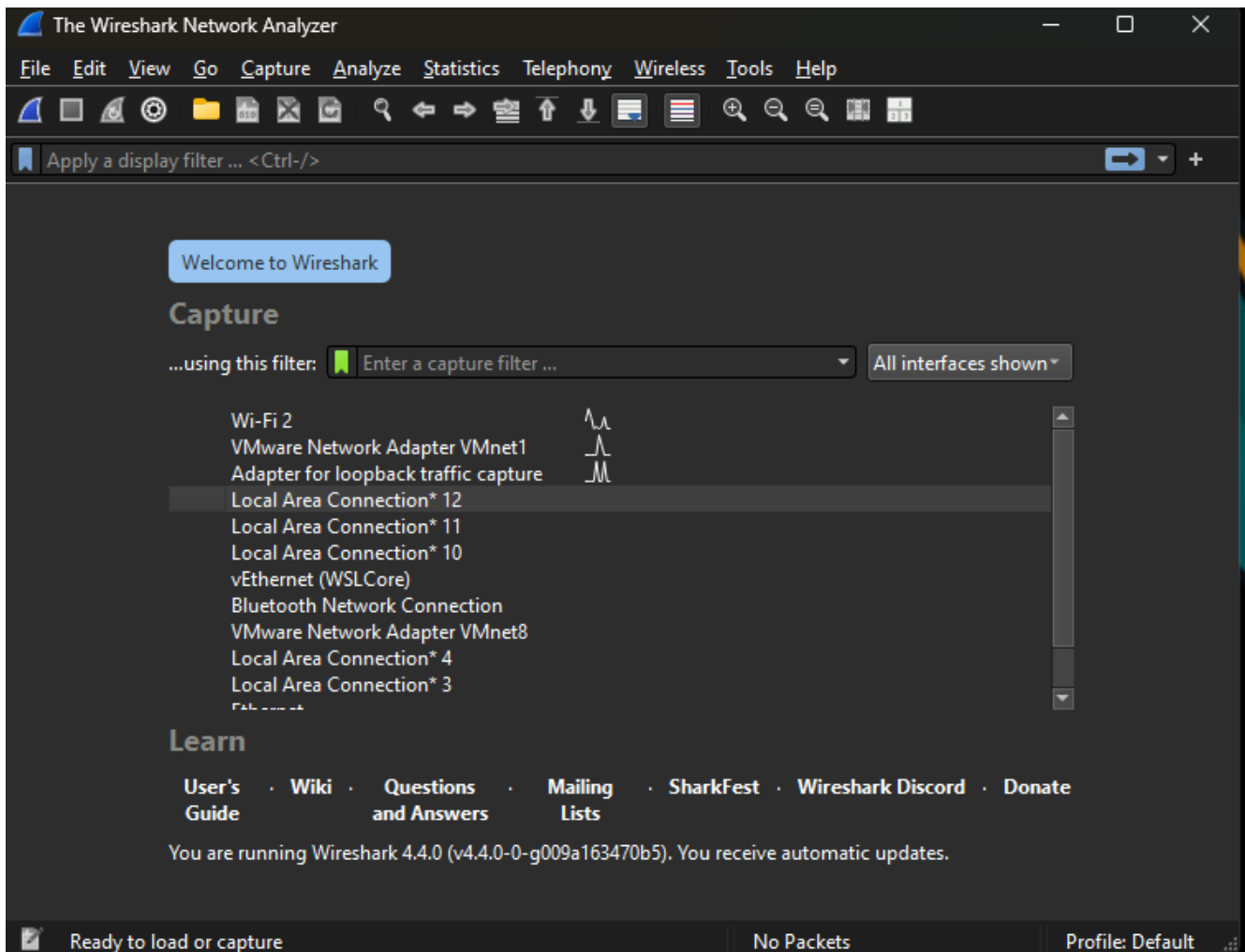
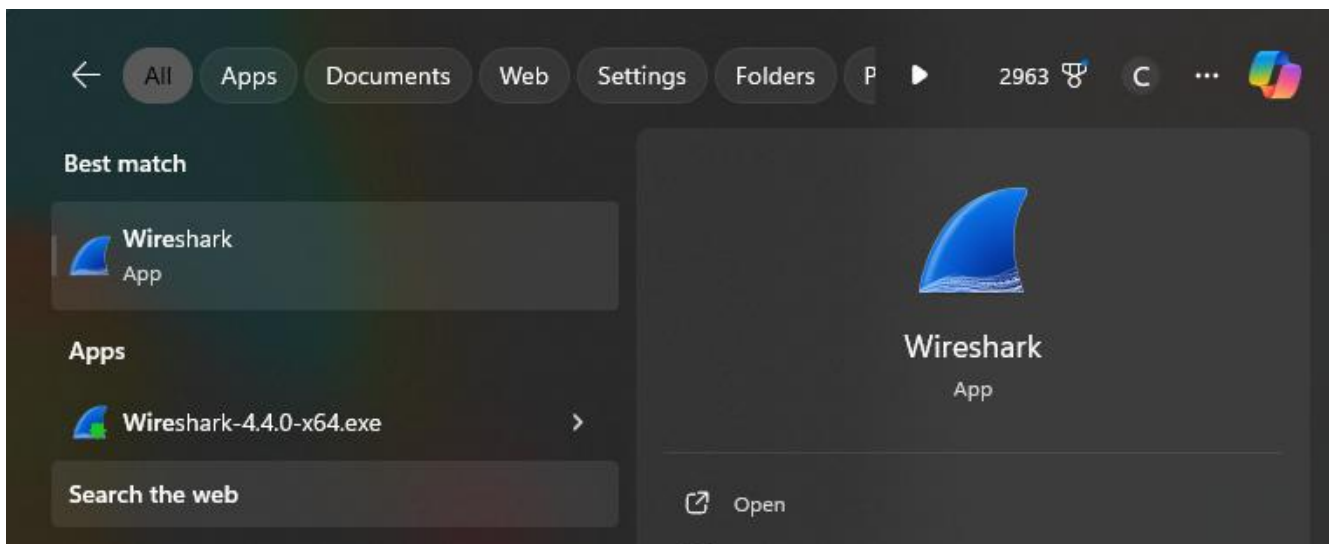
Week 2 Lab Wireshark Familiarity

Step-by-Step Guide: Familiarization with Basic Use of Wireshark and Filtering

Step 1: This should have already been completed. If not please refer to Week 1 Lab.

Step 2: Launch Wireshark: (Screen Shot)

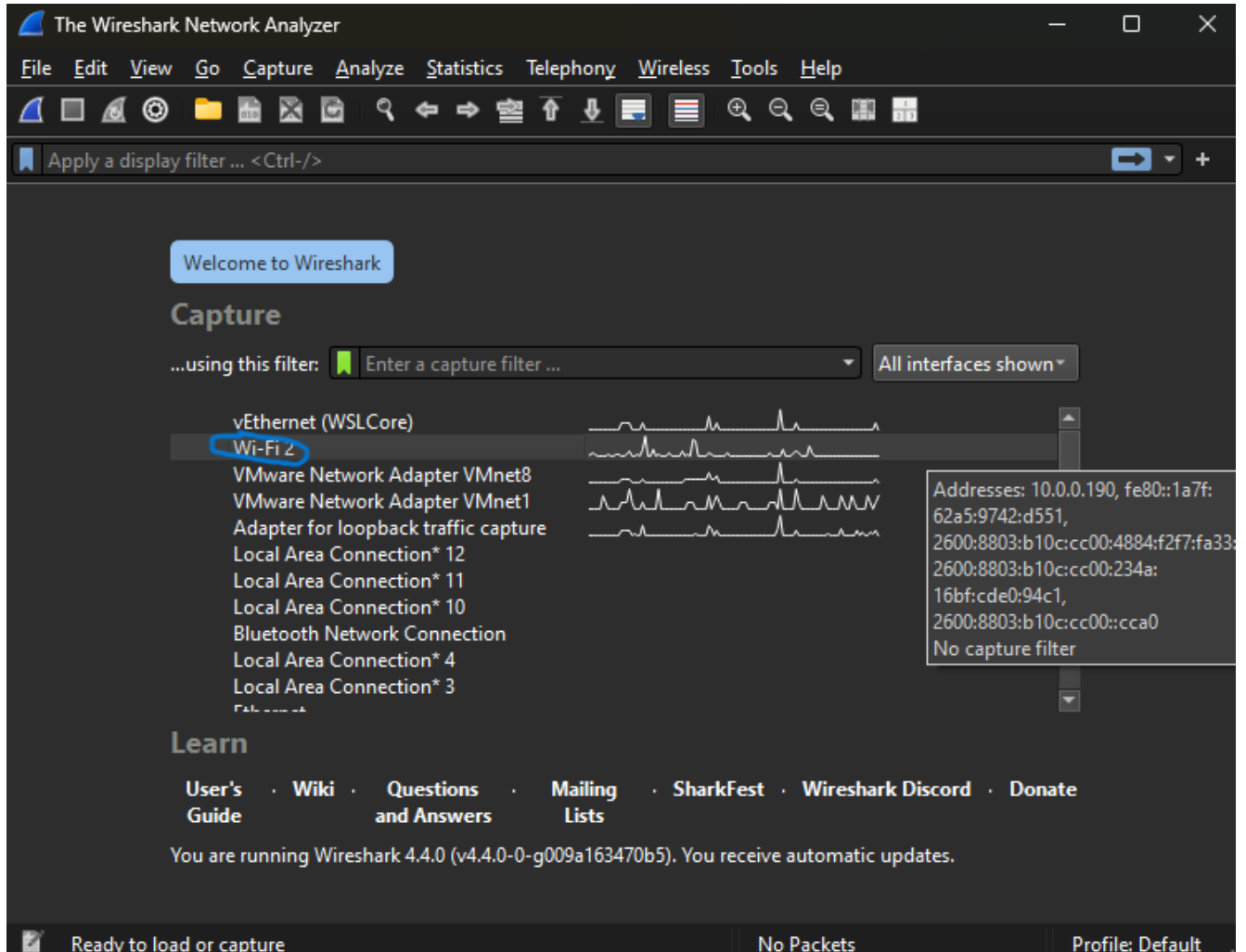
1. Open Wireshark from the desktop shortcut or the Start menu.



Step 3: Select a Network Interface: (Screenshot)

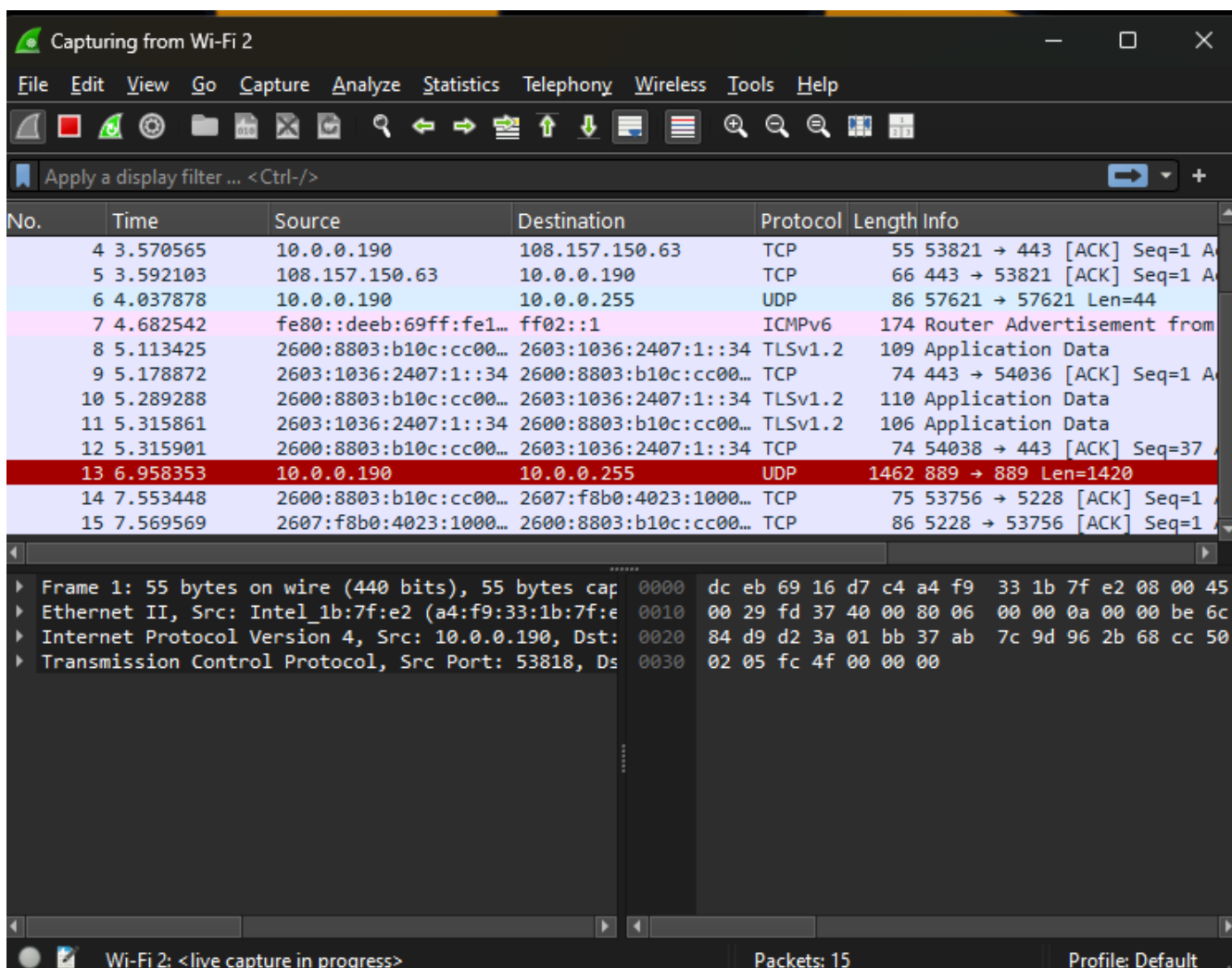
- Upon launching, Wireshark will display a list of available network interfaces. Choose the interface that you want to capture traffic from (e.g., Wi-Fi, Ethernet).

- Click on the desired interface and then click the "Start" button to begin capturing traffic.



Step 4: Capture Packets: (Screenshot)

- Wireshark will start capturing packets in real-time from the selected interface. The captured packets will be displayed in the main window.
- Observe the different columns that show packet information such as source and destination addresses, protocol, length, etc.



Step 5: Filtering HTTP Traffic:(Screenshot)

- In the "Display Filter" field, type "http" and press Enter. This will display only the packets with HTTP traffic.

The screenshot shows the Wireshark interface with a packet capture of HTTP traffic. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like capture, analysis, and display. The packet list on the left shows several HTTP packets, with packet 85 selected. The packet details pane on the right shows the structure of packet 85, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
85	9.2515	10.0.0.190	10.0.0.1	HTTP/X...	354	POST /upnp/control/WANCom
86	9.28617	10.0.0.1	10.0.0.190	HTTP/X...	593	HTTP/1.1 200 OK
96	9.413534	10.0.0.190	10.0.0.1	HTTP/X...	354	POST /upnp/control/WANCom
99	9.420443	10.0.0.1	10.0.0.190	HTTP/X...	593	HTTP/1.1 200 OK
782	24.439736	10.0.0.190	10.0.0.1	HTTP/X...	354	POST /upnp/control/WANCom
785	24.444366	10.0.0.1	10.0.0.190	HTTP/X...	593	HTTP/1.1 200 OK
793	24.450071	10.0.0.190	10.0.0.1	HTTP/X...	354	POST /upnp/control/WANCom
796	24.466794	10.0.0.1	10.0.0.190	HTTP/X...	593	HTTP/1.1 200 OK
869	28.288218	10.0.0.190	10.0.0.165	HTTP	227	GET /dial/dd.xml HTTP/1.1
875	28.395945	10.0.0.165	10.0.0.190	HTTP/X...	1355	HTTP/1.1 200 OK
884	28.684402	10.0.0.190	10.0.0.165	HTTP	245	GET /dial/com.spotify.Spo
888	28.784559	10.0.0.165	10.0.0.190	HTTP	141	HTTP/1.1 404 Not Found

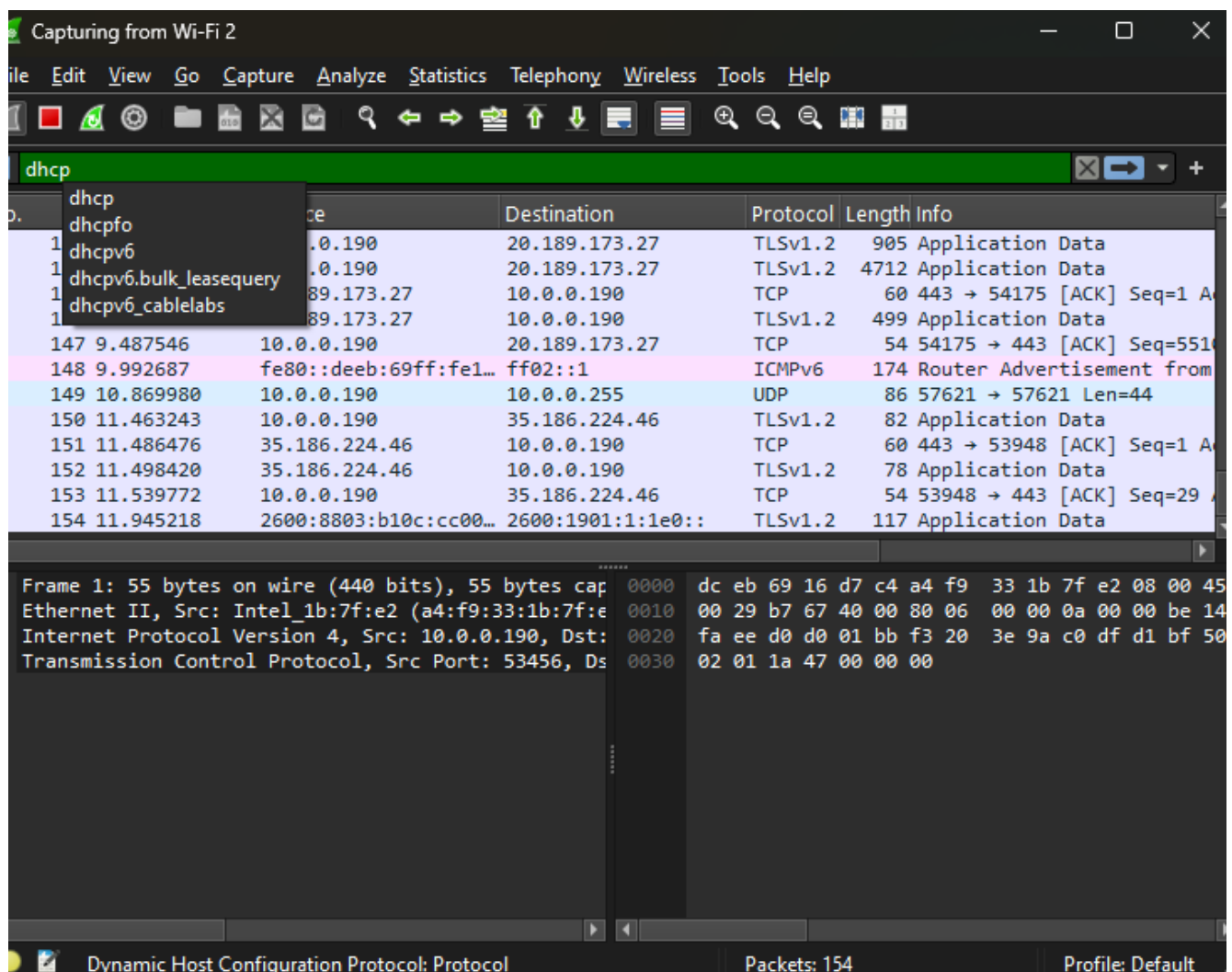
Frame 85: 354 bytes on wire (2832 bits), 354 bytes captured (2832 bits) on interface 0
 Ethernet II, Src: Intel_1b:7f:e2 (a4:f9:33:1b:7f:e2), Dst: 01:00:00:00:00:00
 Internet Protocol Version 4, Src: 10.0.0.190, Dst: 10.0.0.1
 Transmission Control Protocol, Src Port: 54226, Dst Port: 80
 [2 Reassembled TCP Segments (622 bytes): #84(322), #85(300)]
 Hypertext Transfer Protocol

Frame (354 bytes) Reassembled TCP (622 bytes) Decode

Hypertext Transfer Protocol: Protocol Packets: 1383 Displayed: 30 (2.2%) Profile: Default

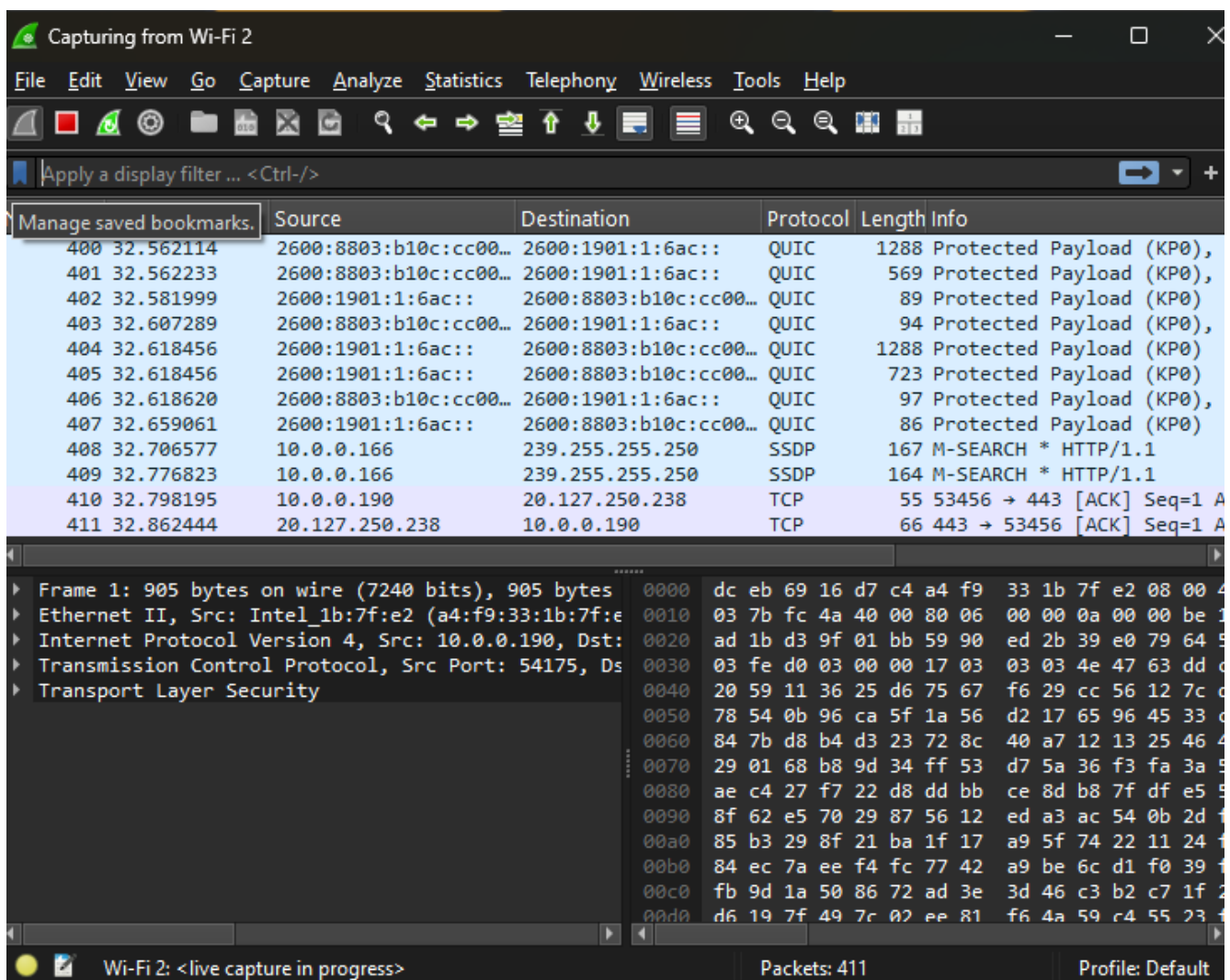
Step 6: Filtering DHCP Traffic:(Screenshot)

- In the "Display Filter" field, type "dhcp" and press Enter. This will display only the packets with DHCP traffic.



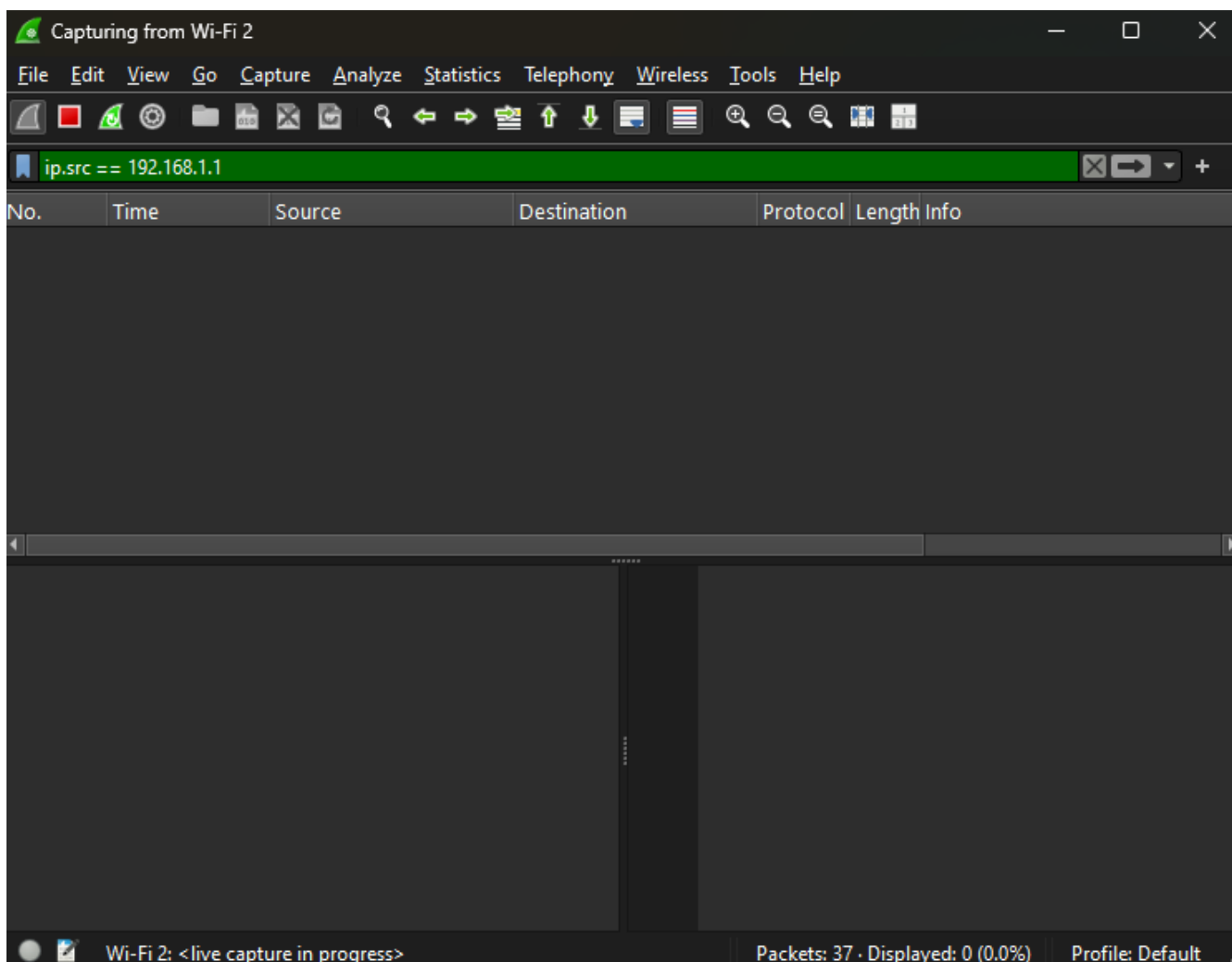
Step 7: Filtering by Port:(Screenshot)

- In the "Display Filter" field, type "port 80" and press Enter. This will display packets using port 80 (default for HTTP).



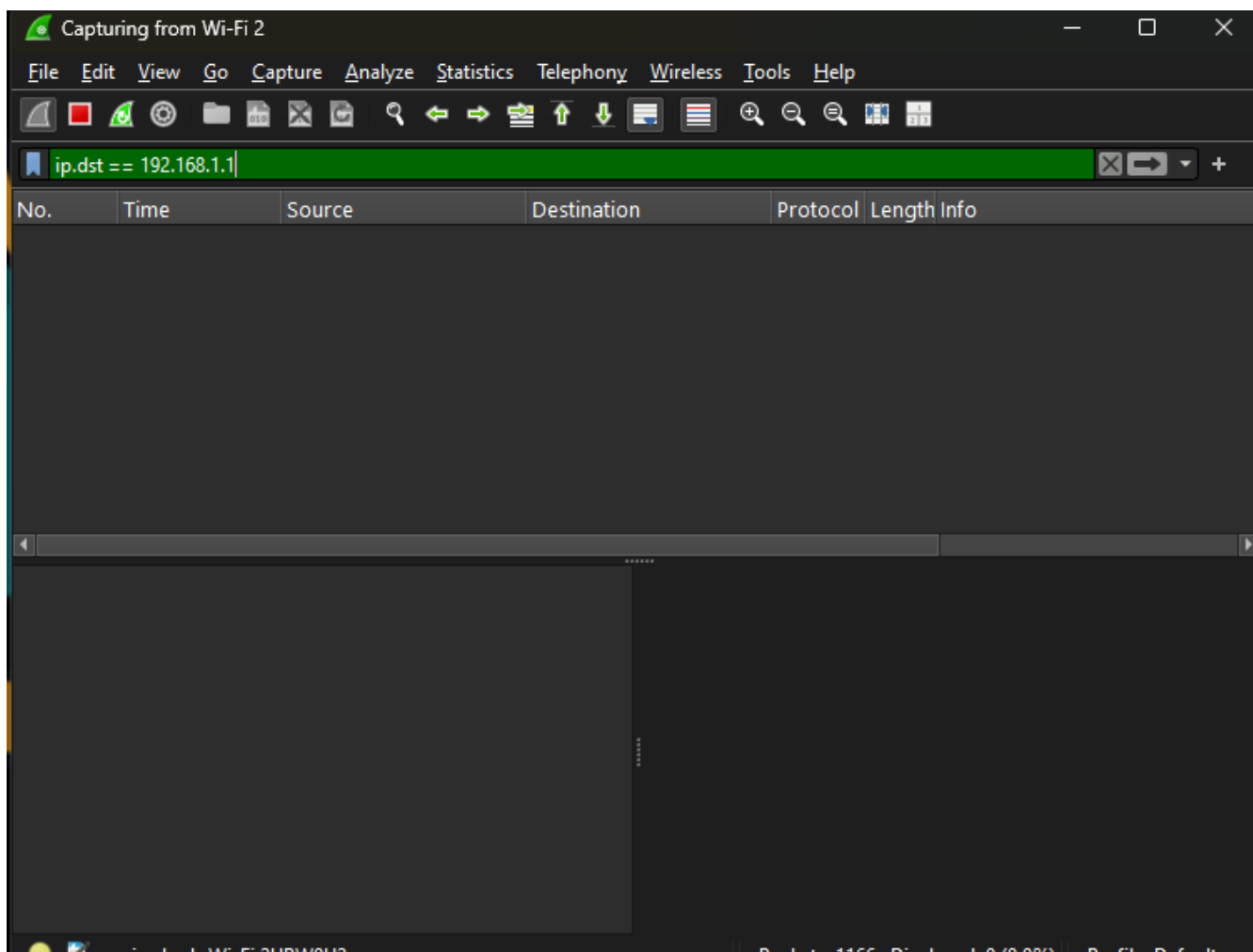
Step 8: Source Filtering:(Screenshot)

- In the "Display Filter" field, type "ip.src == 192.168.1.1" (replace with a specific source IP) and press Enter. This will display packets sent from the specified source IP.



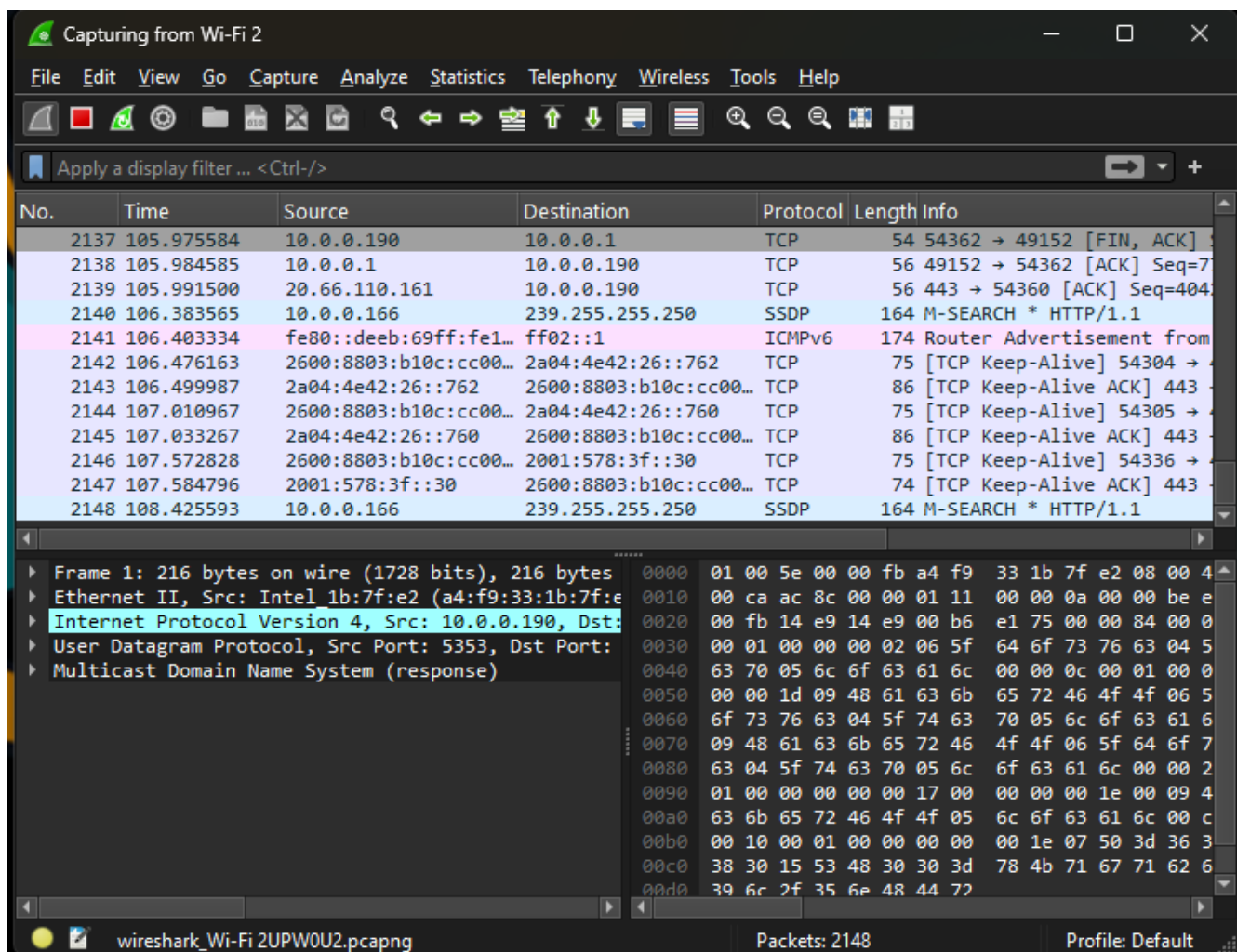
Step 9: Destination Filtering:(Screenshot)

- In the "Display Filter" field, type "ip.dst == 192.168.1.1" (replace with a specific destination IP) and press Enter. This will display packets sent to the specified destination IP.



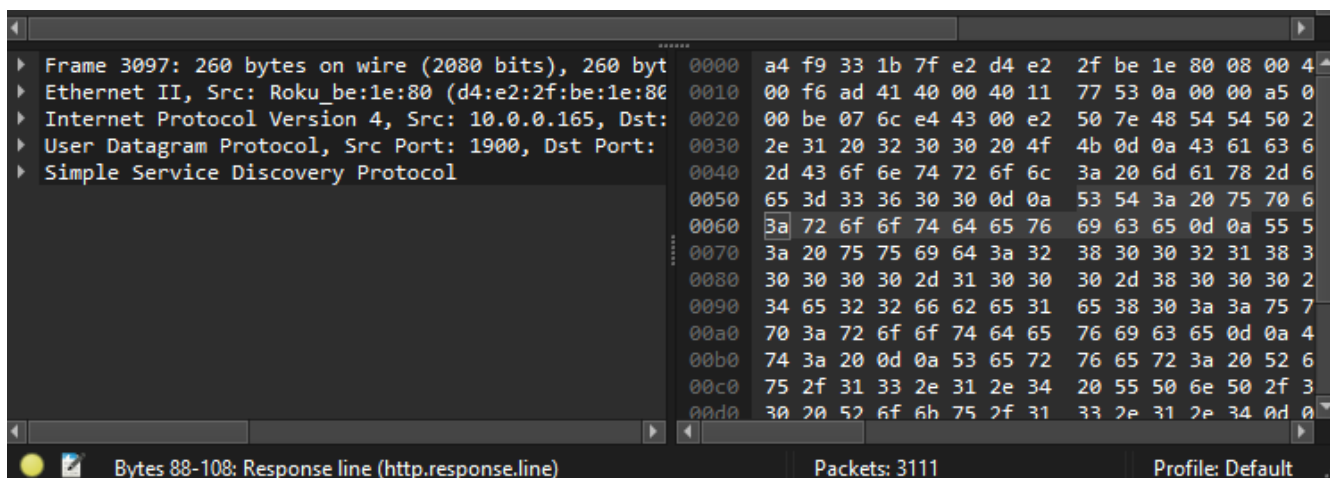
Step 10: Clear Filters:(Screenshot)

- To clear a filter, simply delete the text in the "Display Filter" field and press Enter.



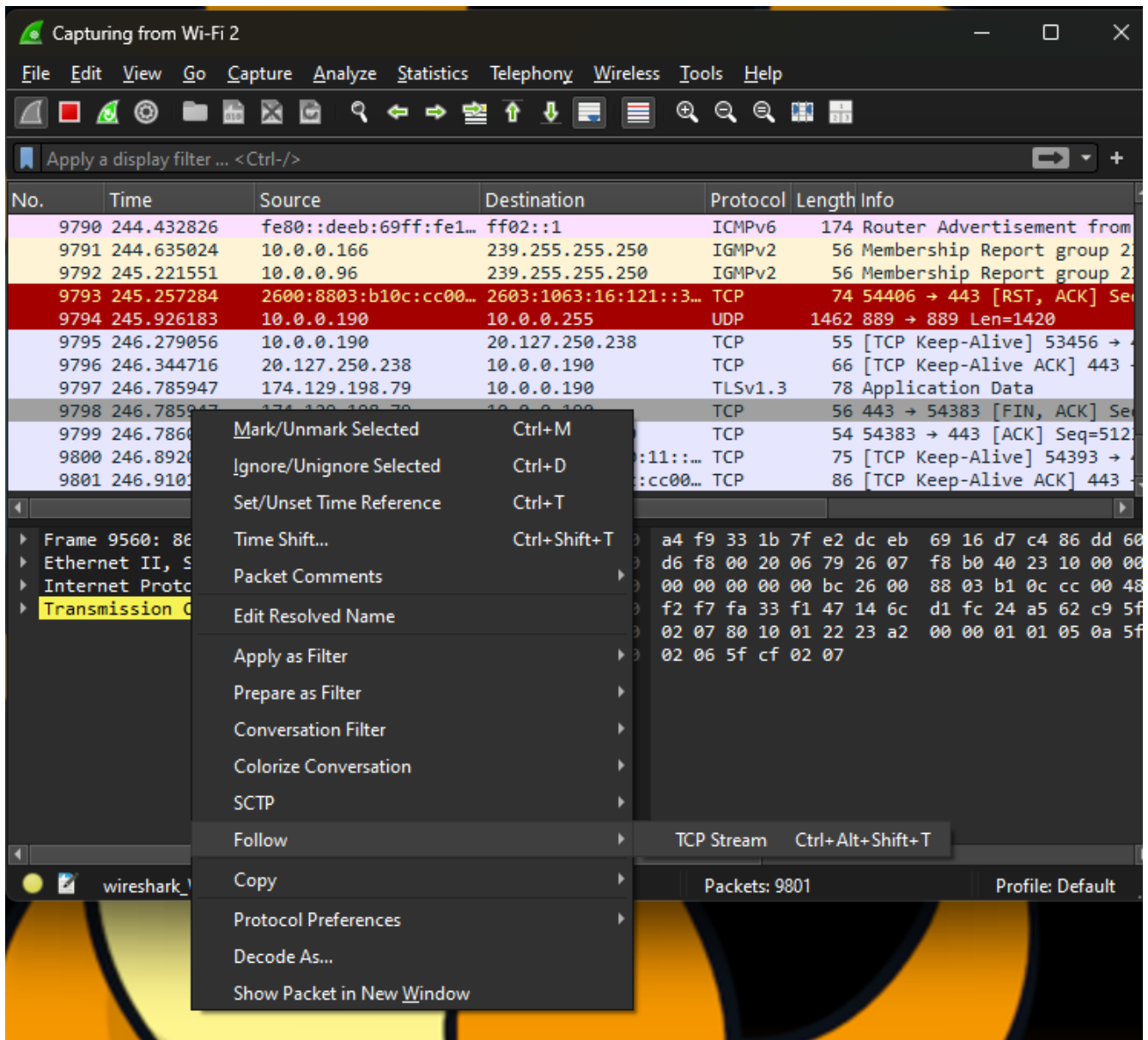
Step 11: Analyze Packet Details:(Screenshot)

- Click on a packet in the list to view its detailed information in the packet details section below. This will display the layers of the packet, from Ethernet to application layer.



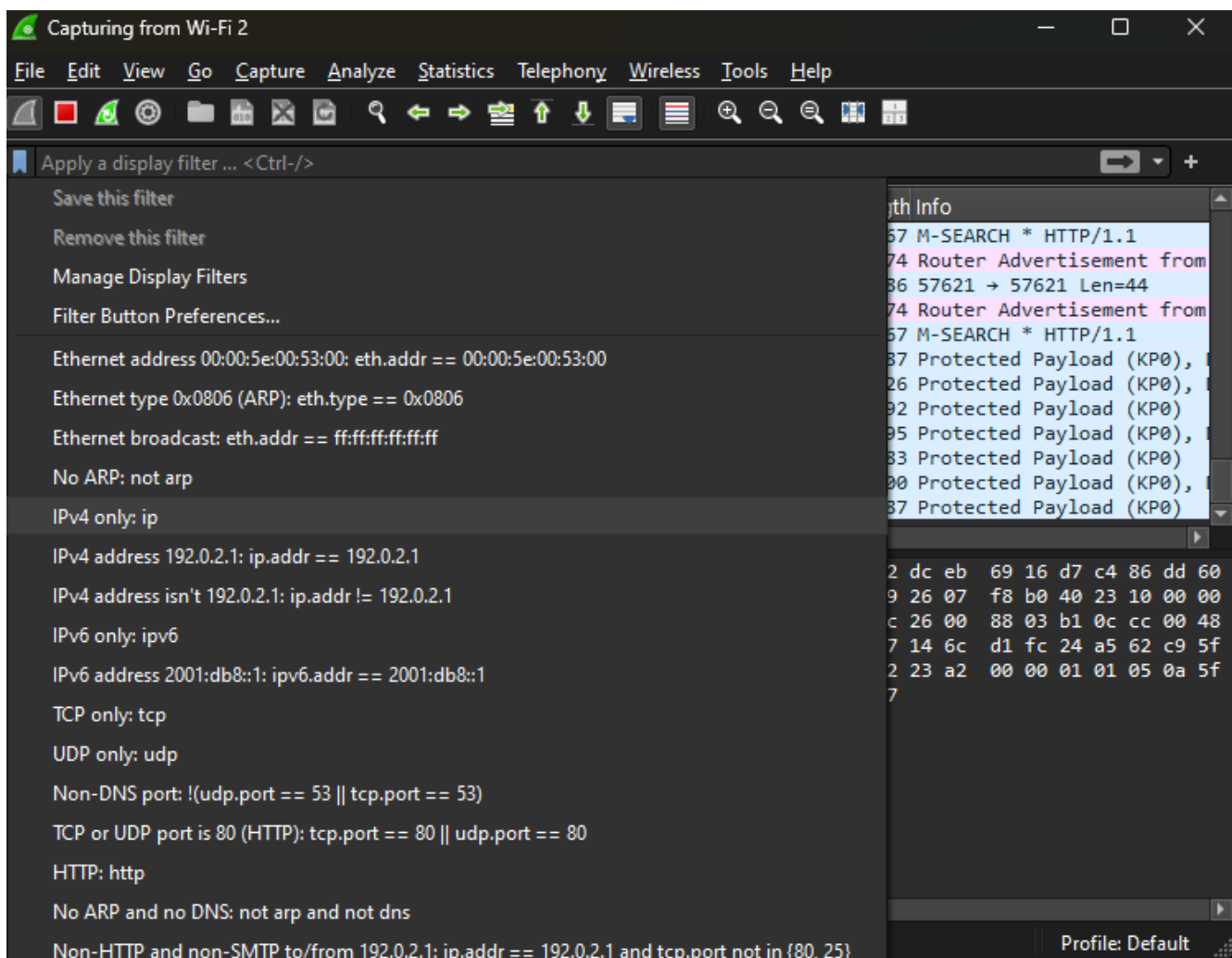
Step 12: Follow TCP Stream and Save Capture:(Screenshot)

- Right-click on a packet and choose "Follow > TCP Stream." This will display the entire conversation between source and destination for that specific TCP stream.
- To save a capture for future analysis, go to "File > Save" and choose a destination. Wireshark will save the captured packets in a ".pcap" file.



Step 13: Practice and Explore:(Screenshot)

- Spend time capturing packets from different scenarios and apply various filters to understand specific traffic patterns.



Step 14: Study Resources: (informational)

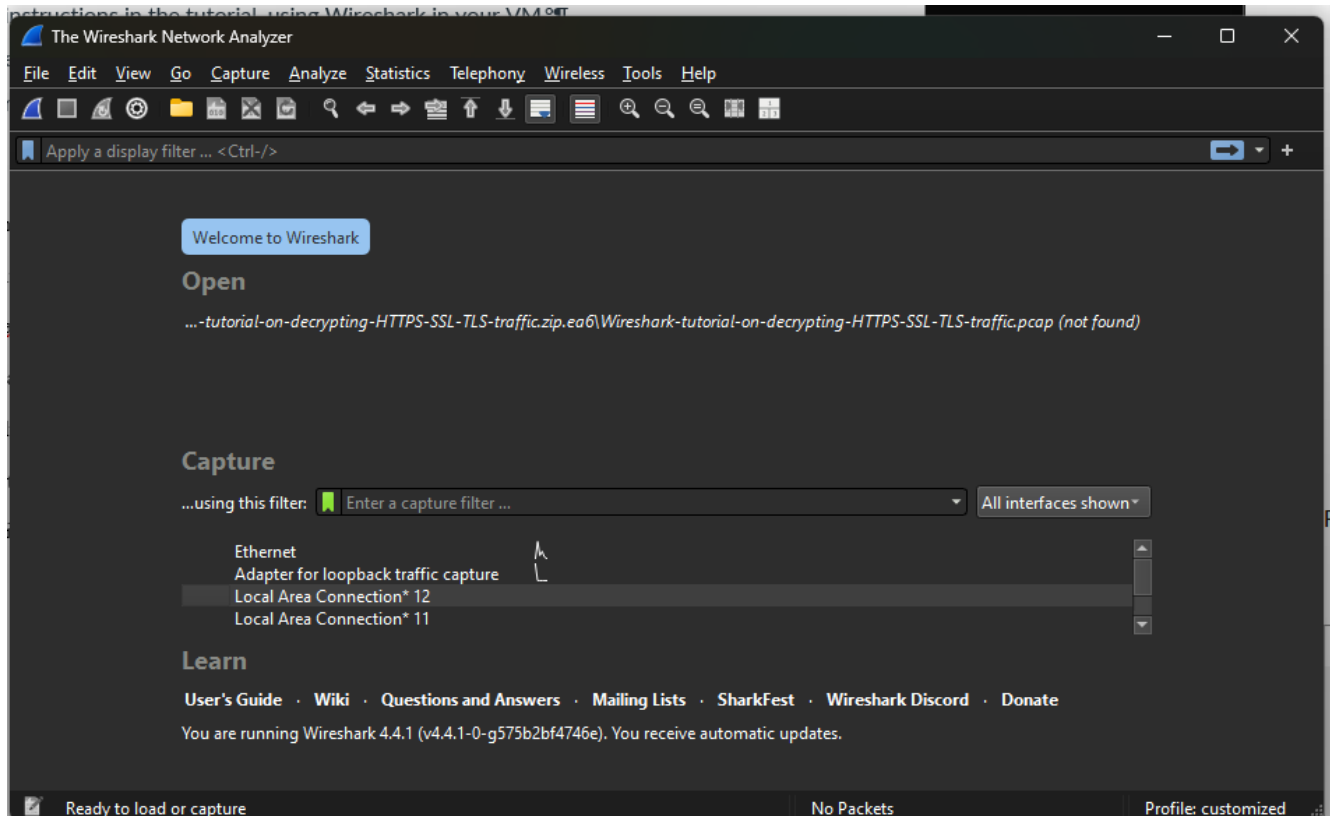
- Utilize Wireshark's official documentation and online tutorials to learn more about its features and functionalities.

Lab Week 8 - Decrypting HTTPS Traffic

Summary

In this lab on decrypting HTTPS traffic, we learned how to look at encrypted network data using Wireshark 3.x. The tutorial is for security professionals who check for suspicious activities in packet captures (pcaps), HTTPS is commonly used by both real websites and malware. By using a special log file that contains encryption keys from the original recording we can decrypt the HTTPS traffic. This

helps us analyze what happens after a malware infection with a specific focus on Dridex malware. We also learned to customize our Wireshark display for clearer viewing of important information. This lab helped us gain important skills for examining encrypted traffic in cybersecurity investigations.



Capturing from vEthernet (WSLCore)

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tcp.stream eq 0

Time	Source	Destination	Protocol	Length	Info
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::fb	MDNS	493	Standard query response 0x0000 PTR, cache flush Hacker...
2024-11-03 21:...	172.17.176.1	224.0.0.251	MDNS	415	Standard query response 0x0000 SRV, cache flush 0 0 76...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::fb	MDNS	435	Standard query response 0x0000 SRV, cache flush 0 0 76...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...

Frame 1: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{967F6F3C...}

Ethernet II, Src: Microsoft_f6:25:a8 (00:15:5d:f6:25:a8), Dst: IPv6mcast_ff:b2:3e:94 (33:33:ff:b2:3e:94)

Internet Protocol Version 6, Src: fe80::d596:f9dd:dd0f:c3f7, Dst: ff02::1:ffb2:3e94

Internet Control Message Protocol v6

0000 33 33 ff b2 3e 94 00 15 5d f6 25 a8

0010 00 00 00 20 3a ff fe 00 00 00 00 00 00 00 00

0020 f9 dd dd 0f c3 f7 ff 00 00 00 00 00 00 00 00

0030 00 01 ff b2 3e 94 87 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 86 69 00 00 00 00 00 00 00 00

0050 00 15 5d f6 25 a8

wireshark_vEthernet (WSLCore)CHAWW2.pcapng Packets: 34 Profile: customized

Capturing from vEthernet (WSLCore)

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(http.request or tls.handshake.type eq 1) and !(ssdp)

Time	Source	Destination	Protocol	Length	Info
2024-11-03 21:...	172.17.176.1	224.0.0.251	MDNS	75	Standard query 0x0000 ANY HackerFOO.local, "QM" questi...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::fb	MDNS	95	Standard query 0x0000 ANY HackerFOO.local, "QM" questi...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::fb	MDNS	133	Standard query response 0x0000 AAAA fe80::d596:f9dd:dd...
2024-11-03 21:...	172.17.176.1	224.0.0.251	MDNS	113	Standard query response 0x0000 AAAA fe80::d596:f9dd:dd...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...
2024-11-03 21:...	fe80::d596:f9dd:dd0...	ff02::1:ffb2:3e94	ICMPv6	86	Neighbor Solicitation for fe80::8669:93ff:feb2:3e94 fr...

Frame 1: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{967F6F3C...}

Ethernet II, Src: Microsoft_f6:25:a8 (00:15:5d:f6:25:a8), Dst: IPv6mcast_ff:b2:3e:94 (33:33:ff:b2:3e:94)

Internet Protocol Version 6, Src: fe80::d596:f9dd:dd0f:c3f7, Dst: ff02::1:ffb2:3e94

Internet Control Message Protocol v6

0000 33 33 ff b2 3e 94 00 15 5d f6 25 a8

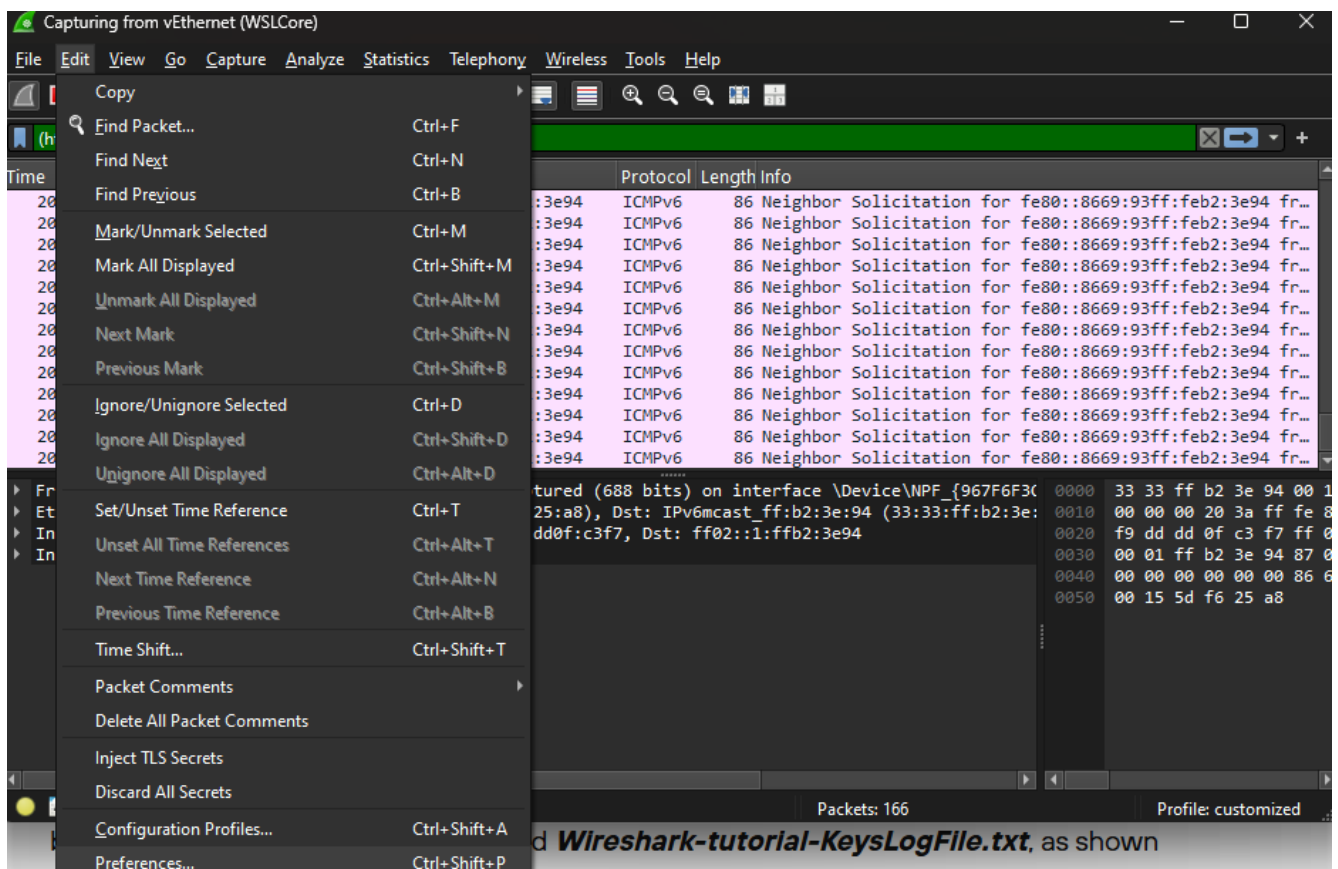
0010 00 00 00 20 3a ff fe 00 00 00 00 00 00 00 00

0020 f9 dd dd 0f c3 f7 ff 00 00 00 00 00 00 00 00

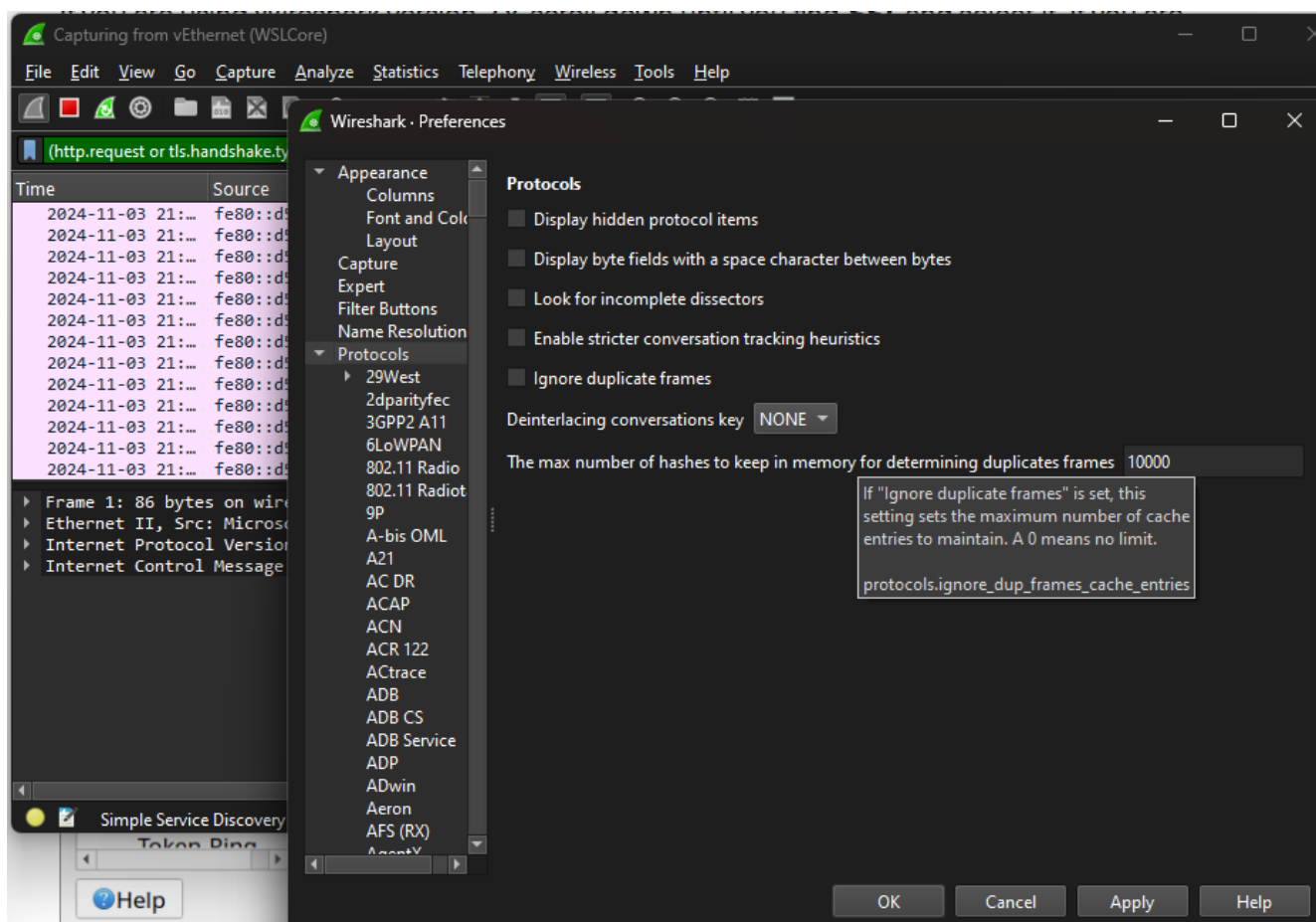
0030 00 01 ff b2 3e 94 87 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 86 69 00 00 00 00 00 00 00 00

0050 00 15 5d f6 25 a8



and **Wireshark-tutorial-KeysLogFile.txt**, as shown



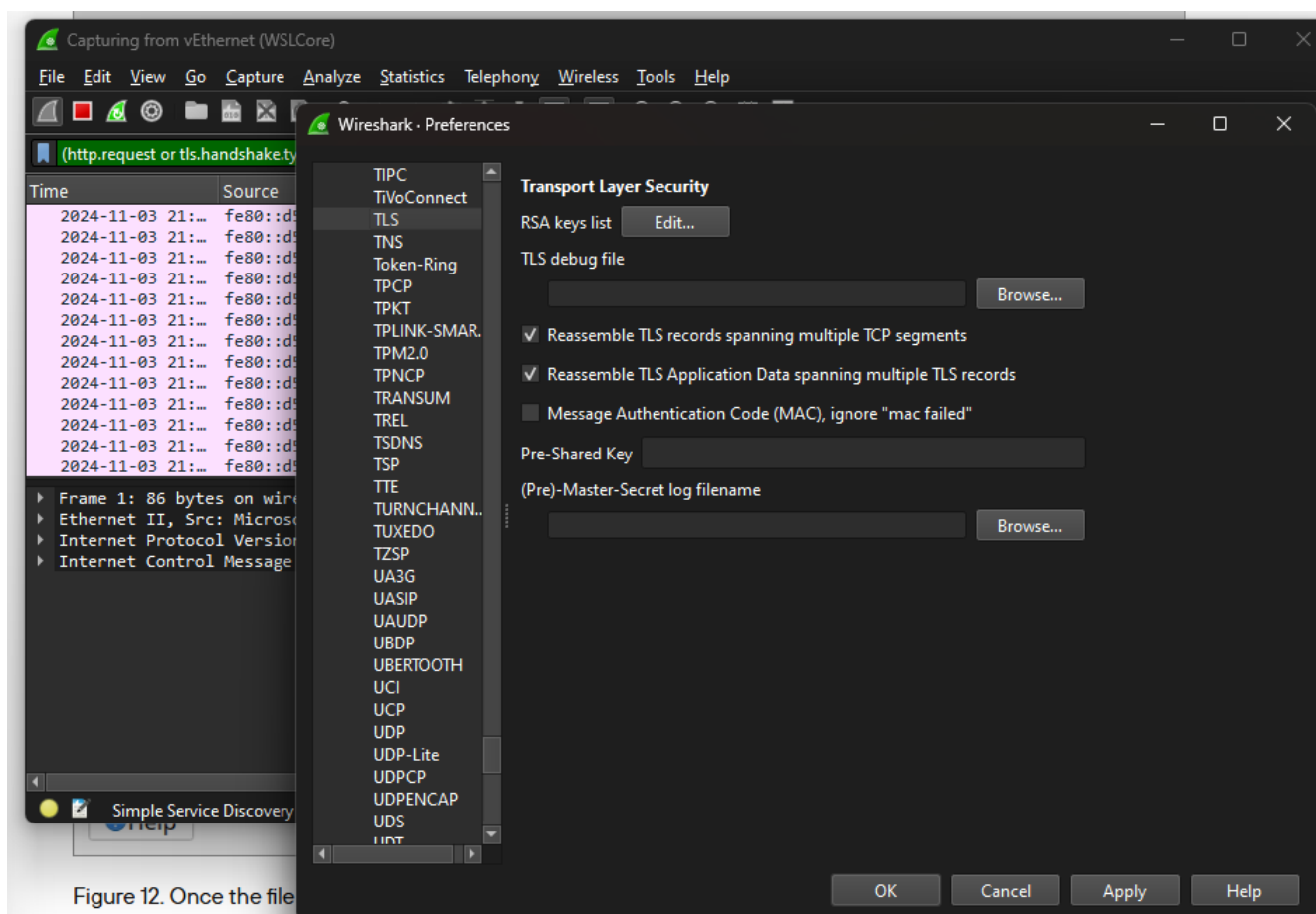


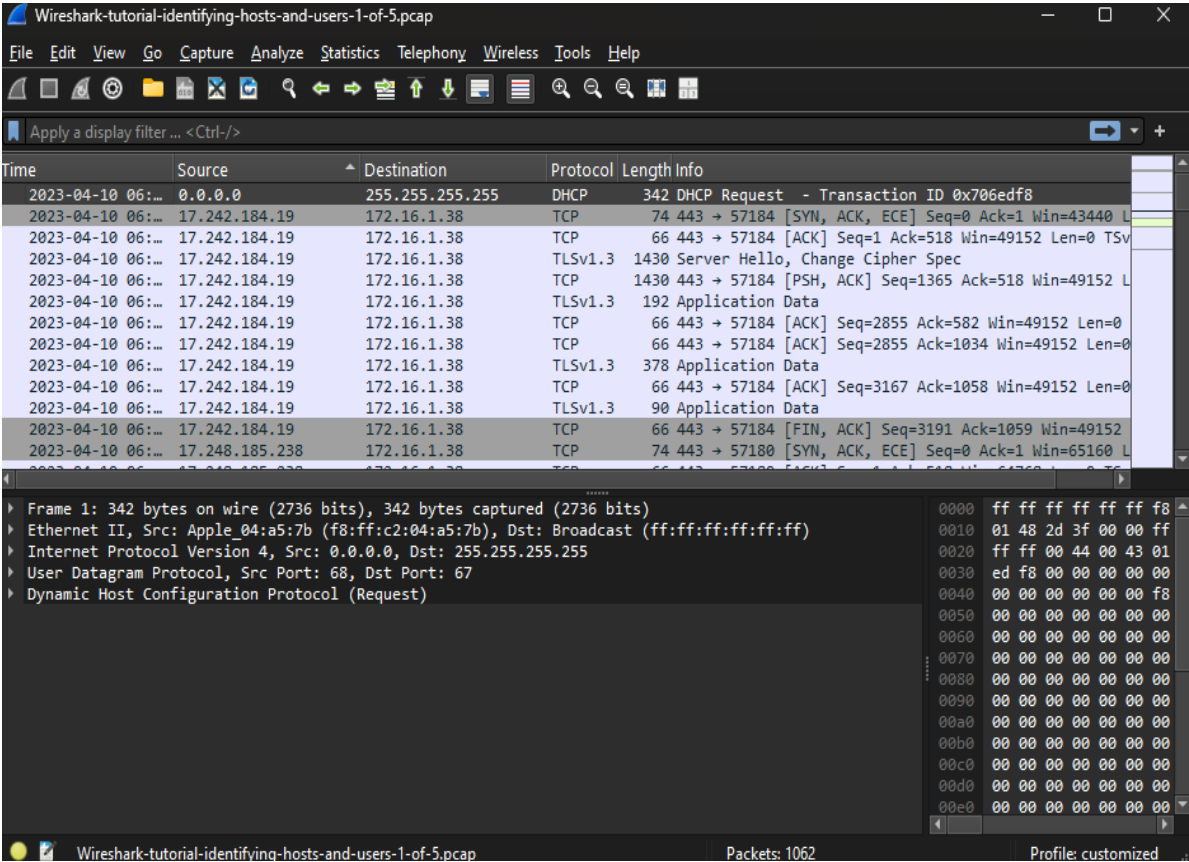
Figure 12. Once the file

Lab Week 9 - Identifying Hosts and Users

Summary

In the Wireshark tutorial on Identifying Hosts and Users, we learned how to capture and look at network traffic to find out which devices and users are on a network. The lab taught us how to use Wireshark to filter and examine data packets to get information about the devices communicating on the network. By looking at details like IP addresses, MAC addresses, and user-agent strings, we could track the activity of certain devices and users. We also learned how to use filters and tools in

Wireshark to focus on traffic from specific devices or apps, which helped us better understand how a network works and how to monitor its security.



Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcpcd

Time	dhcpcd	Destination	Protocol	Length	Info
202	dhcpcd	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x706edf8
202	dhcpcd.bulk_leasereq	172.16.1.38	DHCP	344	DHCP ACK - Transaction ID 0x706edf8

Frame 1: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0
Ethernet II, Src: Apple_04:a5:7b (f8:ff:c2:04:a5:7b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)

Dynamic Host Configuration Protocol: Protocol

Packets: 1062 · Displayed: 2 (0.2%) Profile: customized

Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcpcd

Wireshark · Packet 1 · Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

Transaction ID 0x706edf8

Transaction ID 0x706edf8

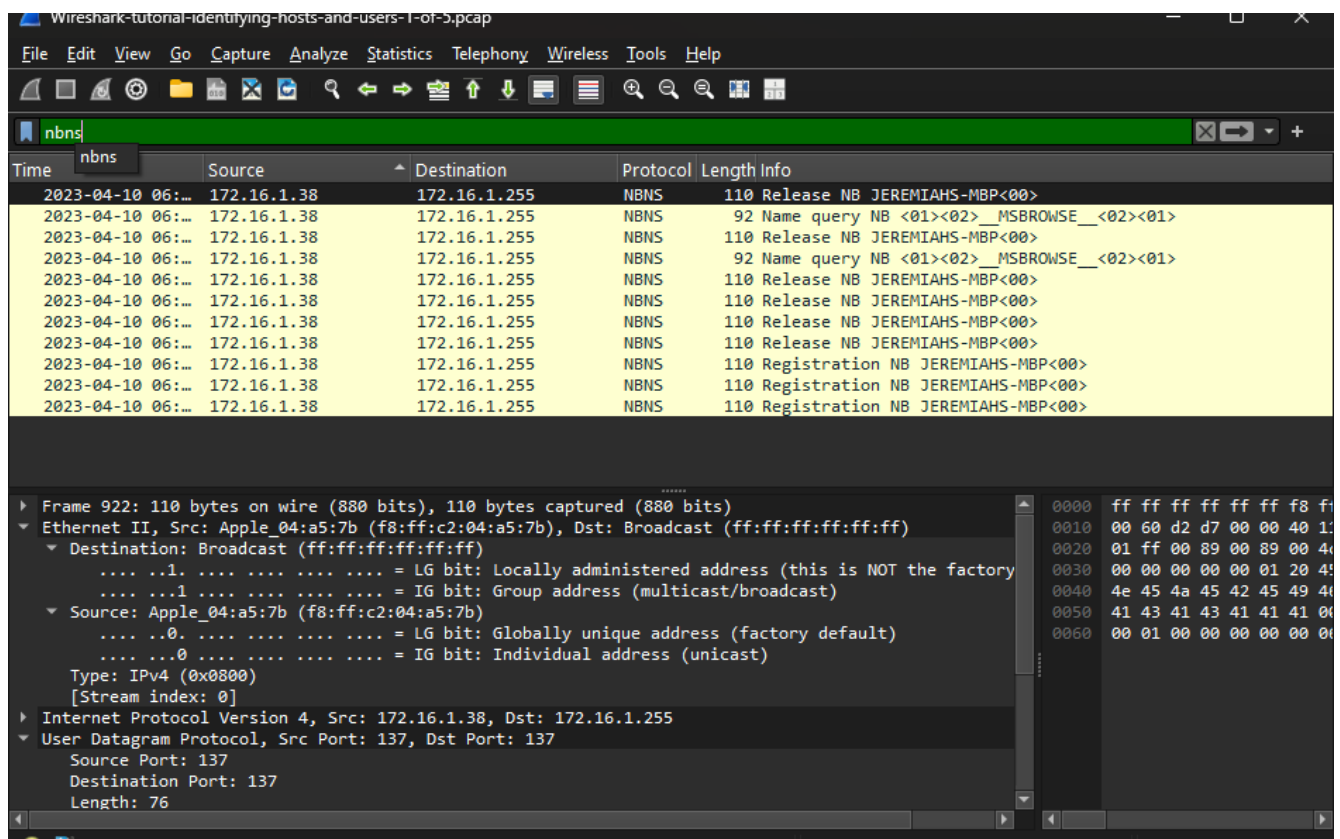
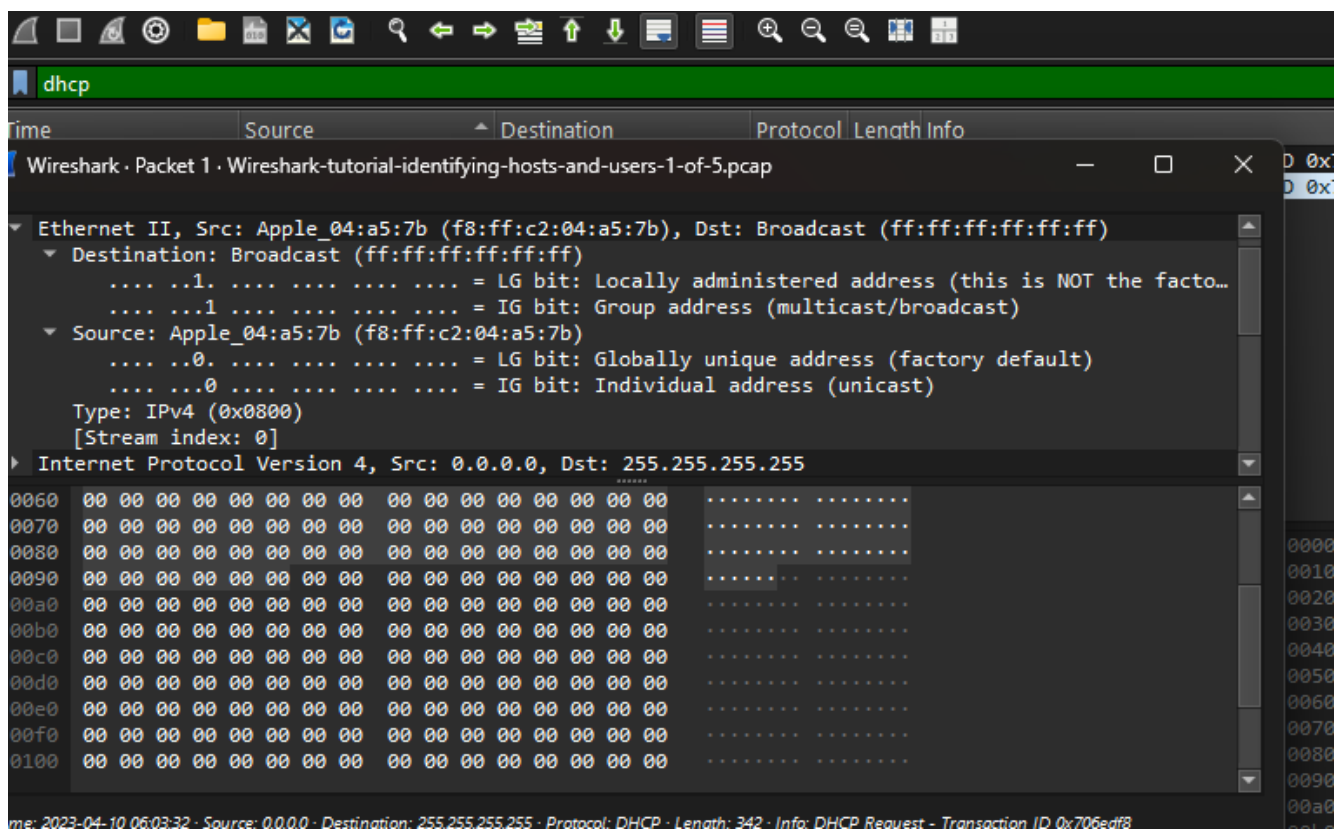
Frame 1: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0
Ethernet II, Src: Apple_04:a5:7b (f8:ff:c2:04:a5:7b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Destination: Broadcast (ff:ff:ff:ff:ff:ff)
Source: Apple_04:a5:7b (f8:ff:c2:04:a5:7b)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Time: 2023-04-10 06:03:32 · Source: 0.0.0.0 · Destination: 255.255.255.255 · Protocol: DHCP · Length: 342 · Info: DHCP Request - Transaction ID 0x706edf8

☒ Show packet bytes Layout: Vertical (Stacked)

Close Help



Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

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nbns

Time	Source	Destination	Protocol	Length	Info
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	92	Name query NB <01><02>__MSBROWSE__<02><01>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	92	Name query NB <01><02>__MSBROWSE__<02><01>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>

Frame 922: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)

Ethernet II, Src: Apple_04:a5:7b (f8:ff:c2:04:a5:7b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

- Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 - ...1. ... = LG bit: Locally administered address (this is NOT the factory)
 - ...1. ... = IG bit: Group address (multicast/broadcast)
- Source: Apple_04:a5:7b (f8:ff:c2:04:a5:7b)
 - ...0. ... = LG bit: Globally unique address (factory default)
 - ...0. ... = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- [Stream index: 0]
- Internet Protocol Version 4, Src: 172.16.1.38, Dst: 172.16.1.255
- User Datagram Protocol, Src Port: 137, Dst Port: 137
 - Source Port: 137
 - Destination Port: 137
 - Length: 76

Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

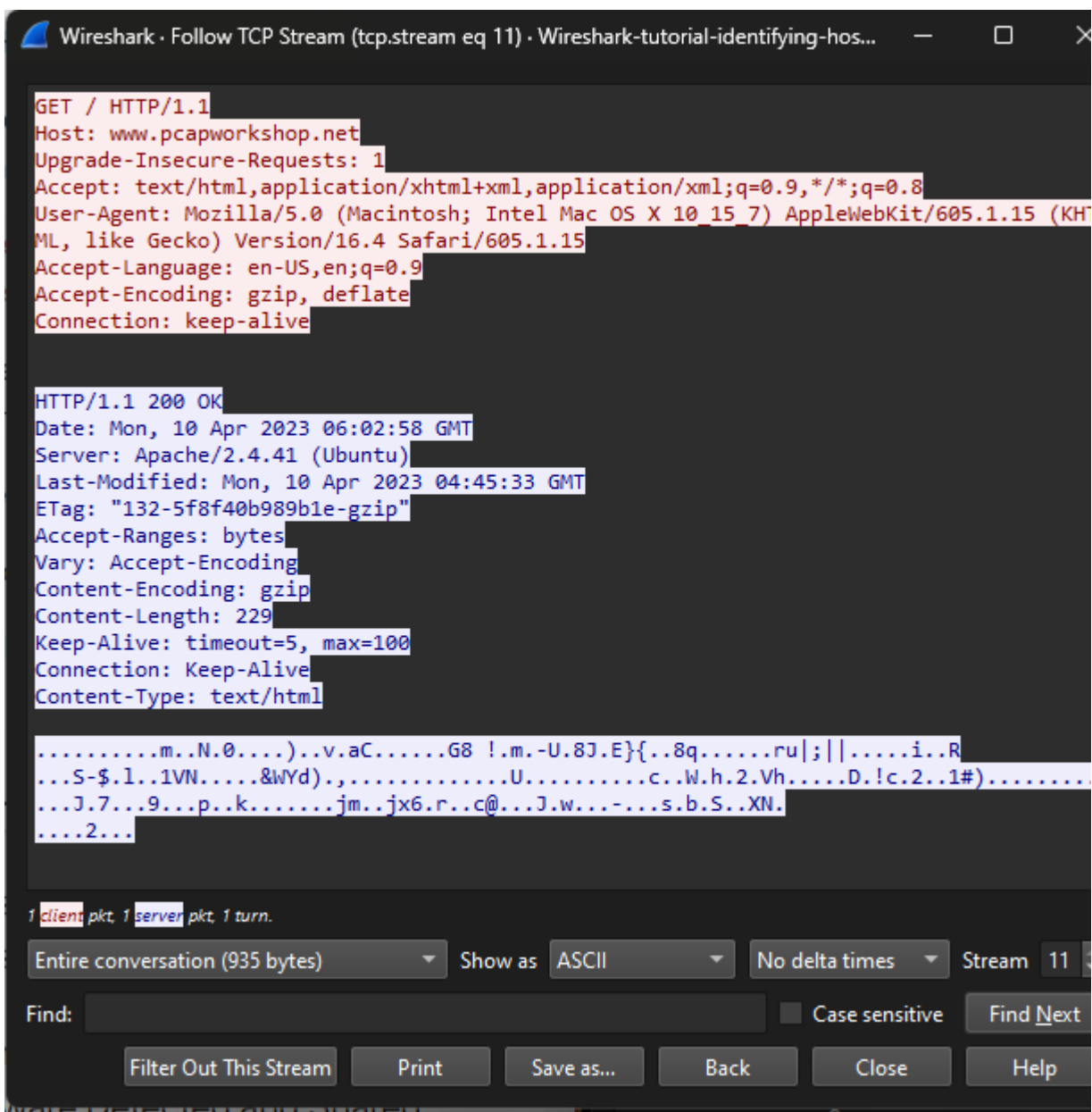
http.accept_language

Time	Source	Destination	Protocol	Length	Info
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	92	Name query NB <01><02>__MSBROWSE__<02><01>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	92	Name query NB <01><02>__MSBROWSE__<02><01>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Release NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>
2023-04-10 06:...	172.16.1.38	172.16.1.255	NBNS	110	Registration NB JEREMIAHS-MBP<00>

Frame 922: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)

Ethernet II, Src: Apple_04:a5:7b (f8:ff:c2:04:a5:7b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

- Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 - ...1. ... = LG bit: Locally administered address (this is NOT the factory)
 - ...1. ... = IG bit: Group address (multicast/broadcast)
- Source: Apple_04:a5:7b (f8:ff:c2:04:a5:7b)
 - ...0. ... = LG bit: Globally unique address (factory default)
 - ...0. ... = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- [Stream index: 0]
- Internet Protocol Version 4, Src: 172.16.1.38, Dst: 172.16.1.255
- User Datagram Protocol, Src Port: 137, Dst Port: 137
 - Source Port: 137
 - Destination Port: 137
 - Length: 76



Wireshark-tutorial-identifying-hosts-and-users-1-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(http.request or tls.handshake.type eq 1) and !ssdp

Time	Source	Destination	Protocol	Length	Info
2023-04-10 06:...	172.16.1.38	17.253.127.202	HTTP	198	GET /hotspot-detect.html HTTP/1.0
2023-04-10 06:...	172.16.1.38	96.7.172.24	TLSv1.3	583	Client Hello (SNI=apps.mzstatic.com)
2023-04-10 06:...	172.16.1.38	17.253.127.203	TLSv1.3	583	Client Hello (SNI=ipcdn.apple.com)
2023-04-10 06:...	172.16.1.38	17.253.127.203	TLSv1.3	583	Client Hello (SNI=ipcdn.apple.com)
2023-04-10 06:...	172.16.1.38	17.253.127.214	TLSv1.3	583	Client Hello (SNI=ipcdn.apple.com)
2023-04-10 06:...	172.16.1.38	17.57.144.88	TLSv1.3	583	Client Hello (SNI=courier.push.apple.com)
2023-04-10 06:...	172.16.1.38	17.248.185.238	TLSv1.2	583	Client Hello (SNI=p31-fmfmobile.icloud.com)
2023-04-10 06:...	172.16.1.38	23.40.180.144	TLSv1.3	583	Client Hello (SNI=gspe1-ssl.ls.apple.com)
2023-04-10 06:...	172.16.1.38	17.248.241.231	TLSv1.3	583	Client Hello (SNI=gateway.icloud.com)
2023-04-10 06:...	172.16.1.38	17.242.184.19	TLSv1.3	583	Client Hello (SNI=gs-loc.apple.com)
2023-04-10 06:...			HTTP	435	GET / HTTP/1.1
2023-04-10 06:...			HTTP	395	GET /favicon.ico HTTP/1.1

Mark/Unmark Selected Ctrl+M

Ignore/Unignore Selected Ctrl+D

Set/Unset Time Reference Ctrl+T

Time Shift... Ctrl+Shift+T

Packet Comments

Edit Resolved Name

Apply as Filter

Prepare as Filter

Conversation Filter

Colorize Conversation

SCTP

Follow

Copy

Protocol Preferences

Decode As...

Show Packet in New Window

[Stream index: 0]

Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Time: 2023-04-10 06:03:32 · Source: 0.0.0.0 · Destination: 255.255.255.255 · Protocol: DHCP · Length: 342 · Info: DHCP Request - Transaction ID 0x706edf8

✓ Show packet bytes Layout: Vertical (Stacked)

Wireshark - Follow TCP Stream (tcp.stream eq 11) - Wireshark-tutorial-identifying-hos...

GET / HTTP/1.1
Host: www.pcapworkshop.net
Upgrade-Insecure-Requests: 1
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/16.4 Safari/605.1.15
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate
Connection: keep-alive

HTTP/1.1 200 OK
Date: Mon, 10 Apr 2023 06:02:58 GMT
Server: Apache/2.4.41 (Ubuntu)
Last-Modified: Mon, 10 Apr 2023 04:45:33 GMT
ETag: "132-5f8f40b989b1e-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 229
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html

.....m..N.0....).v.v.c.....G8 !.m.-U.8J.E}{.8q.....ru;||.....i..R
...S-\$..1..1VN.....&MYd).....U.....c..W.h.2.Vh.....D.l.c.2..1#).....
...J.7...9...p..k.....jm...jx6.r..c@...J.w...-...s.b.S..XN.
...2...

1 client pkt, 1 server pkt, 1 turn.

Entire conversation (935 bytes) Show as ASCII No delta times Stream 11

Find: Case sensitive Find Next

Wireless Tools Help

Protocol	Length	Info
TCP	78	57186 → 80 [SYN, ECE, Chr] Seq=0 Win=65535 Len=0 MSS=146...
TCP	66	57186 → 80 [ACK] Seq=1 Ack=1 Win=131456 Len=0 TSval=1408...
HTTP	435	GET / HTTP/1.1
TCP	66	57186 → 80 [ACK] Seq=370 Ack=567 Win=130880 Len=0 TSval=...
TCP	74	80 → 57186 [SYN, ACK, ECE] Seq=0 Ack=1 Win=65160 Len=0 ML...
TCP	66	80 → 57186 [ACK] Seq=1 Ack=370 Win=64896 Len=0 TSval=171...
HTTP	632	HTTP/1.1 200 OK (text/html)

captured (3480 bits)
..., Dst: Cisco_7b:85:31 (00:08:32:7b:85:31)

ly unique address (factory default)
dual address (unicast)

ly unique address (factory default)
dual address (unicast)

203.161.53.240
Port: 80, Seq: 1, Ack: 1, Len: 369

Packets: 1062 · Displayed: 7 (0.7%) Profile: customized

Wireshark-tutorial-identifying-nodes-and-users-3-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

kerberos.CNameString

Time	Source	Destination	Protocol	Length	CNameString	Info
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	382	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	419	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	268	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	382	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	439	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	419	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	383	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	439	DESKTOP-HSJD5R8\$	TGS-REP

pvno: 5
msg-type: krb-as-req (10)
padata: 1 item
req-body
padding: 0
kdc-options: 40810010
cname
name-type: kRB5-NT-PRINCIPAL (1)
cname-string: 1 item
CNameString (kerberos) realm: pcap
sname
till: Sep :
rtime: Sep :
nonce: 821:
etype: 6 i

Expand Subtrees
Collapse Subtrees
Expand All
Collapse All
Apply as Column Ctrl+Shift+I

Time
Time

Packets: 12329 · Displayed: 37 (0.3%) Profile: customized

Wireshark-tutorial-identifying-hosts-and-users-3-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

kerberos.CNameString

Time	Source	Destination	Protocol	Length	CNameString	Info
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	382	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	419	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	268	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	382	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	439	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	419	DESKTOP-HSJD5R8\$	TGS-REP
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	302	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	383	desktop-hsjd5r8\$	AS-REQ
2023-03-10 19:...	172.16.1.16	172.16.1.141	KRB5	405	DESKTOP-HSJD5R8\$	AS-REP

pvno: 5
msg-type: krb-as-req (10)
padata: 1 item
req-body
padding: 0
kdc-options: 40810010
cname
name-type: kRB5-NT-PRINCIPAL (1)
cname-string: 1 item
CName
realm: pcap
sname
till: Sep
rtime: Sep
nonce: 821

Expand Subtrees
Collapse Subtrees
Expand All

Time
Time

0040 a1 03 02 01 05 a2 03
0050 a1 04 02 02 00 80 a2
0060 ff a4 81 ca 30 81 c7
0070 a1 1d 30 1b a0 03 02
0080 65 73 6b 74 6f 70 2d
0090 12 1b 10 70 63 61 70
00a0 6e 65 74 a3 25 30 23
00b0 1b 06 6b 72 62 74 67
00c0 72 6b 73 68 6f 70 2e
00d0 30 30 30 39 31 33 30
00e0 0f 32 31 30 30 30 39
00f0 a7 06 02 04 30 f3 64
0100 01 11 02 01 17 02 01
0110 1d 30 1b 30 19 a0 03
0120 53 4b 54 4f 50 2d 48

Wireshark-tutorial-identifying-hosts-and-users-3-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ldap contains "CN=Users"

Time	Source	Destination	Protocol	Length	CNameString	Info
2023-03-10 19:...	172.16.1.141	172.16.1.16	LDAP	212		SASL GSS-API Integrity: searchRequest(4)
2023-03-10 19:...	172.16.1.16	172.16.1.141	LDAP	234		SASL GSS-API Integrity: searchResEntry(4)
2023-03-10 19:...	172.16.1.141	172.16.1.16	LDAP	212		SASL GSS-API Integrity: searchRequest(9)
2023-03-10 19:...	172.16.1.16	172.16.1.141	LDAP	234		SASL GSS-API Integrity: searchResEntry(9)

Frame 6030: 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits)
Ethernet II, Src: 00:51:8a:93:d2:f6 (00:51:8a:93:d2:f6), Dst: Dell_27:b0:f9 (00:1e:4f:27:b0:f9)
Internet Protocol Version 4, Src: 172.16.1.141, Dst: 172.16.1.16
Transmission Control Protocol, Src Port: 65483, Dst Port: 389, Seq: 2409, Ack: 3022, Len: 158
Lightweight Directory Access Protocol

0000 00 1e 4f 27 b0 f9 00 51
0010 00 c6 6d f7 40 00 80 00
0020 01 10 ff cb 01 85 d5 a5
0030 04 01 93 00 00 00 00 00
0040 00 0c 00 00 00 00 47 b0
0050 fa 00 1d 7f 34 b3 30 84
0060 84 00 00 00 6f 04 30 43
0070 63 43 6f 6c 6c 75 6d 20
0080 2c 44 43 3d 70 63 61 70
0090 2c 44 43 3d 6e 65 74 00
00a0 02 01 00 01 01 00 a3 84
00b0 6a 65 63 74 43 6c 61 73
00c0 84 00 00 0f 04 09 60
00d0 04 02 73 6e

Wireshark-tutorial-identifying-hosts-and-users-3-of-5.pcap

Packets: 12329 · Displayed: 4 (0.0%)

Profile: customized

Wireshark-tutorial-identifying-hosts-and-users-3-of-5.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip contains "DESKTOP-"

Time	Source	Destination	Protocol	Length	CNameString	Info
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	380	rene.mccollum	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	291	rene.mccollum	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.16	KRB5	371	rene.mccollum	AS-REQ
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.16	DCERPC	206		Alter_context: call_id: 18, Fragme
2023-03-10 19:...	172.16.1.141	172.16.1.16	DCERPC	206		Alter_context: call_id: 19, Fragme
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5
2023-03-10 19:...	172.16.1.141	172.16.1.16	CLDAP	269		searchRequest(20) "<ROOT>" baseObj
2023-03-10 19:...	172.16.1.141	172.16.1.255	BROWSER	228		Request Announcement DESKTOP-HSJD5

Frame 4623: 228 bytes on wire (1824 bits), 228 bytes captured (1824 bits)

Ethernet II, Src: 00:51:8a:93:d2:f6 (00:51:8a:93:d2:f6), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Internet Protocol Version 4, Src: 172.16.1.141, Dst: 172.16.1.255

User Datagram Protocol, Src Port: 138, Dst Port: 138

NetBIOS Datagram Service

SMB (Server Message Block Protocol)

SMB MailSlot Protocol

Microsoft Windows Browser Protocol

0000 ff ff ff ff ff ff 00 5

0010 00 d6 bc 83 00 00 80 1

0020 01 ff 00 8a 00 8a 00 c

0030 01 8d 00 8a 00 ac 00 0

0040 4c 46 45 45 50 46 41 4

0050 45 44 46 46 43 44 49 4

0060 42 46 41 46 48 45 50 4

0070 50 46 41 43 41 43 41 4

0080 25 00 00 00 00 00 00 0

0090 00 00 00 00 00 00 00 0

00a0 00 00 00 00 00 00 00 0

00b0 00 00 00 12 00 56 00 0

00c0 00 5c 4d 41 49 4c 53 4

00d0 45 00 02 00 44 45 53 4

00e0 35 52 38 00